

Detecting and Assessing Pollution Events from Wildfires Using Remote Sensing and Meteorological Data: A Data Science Approach

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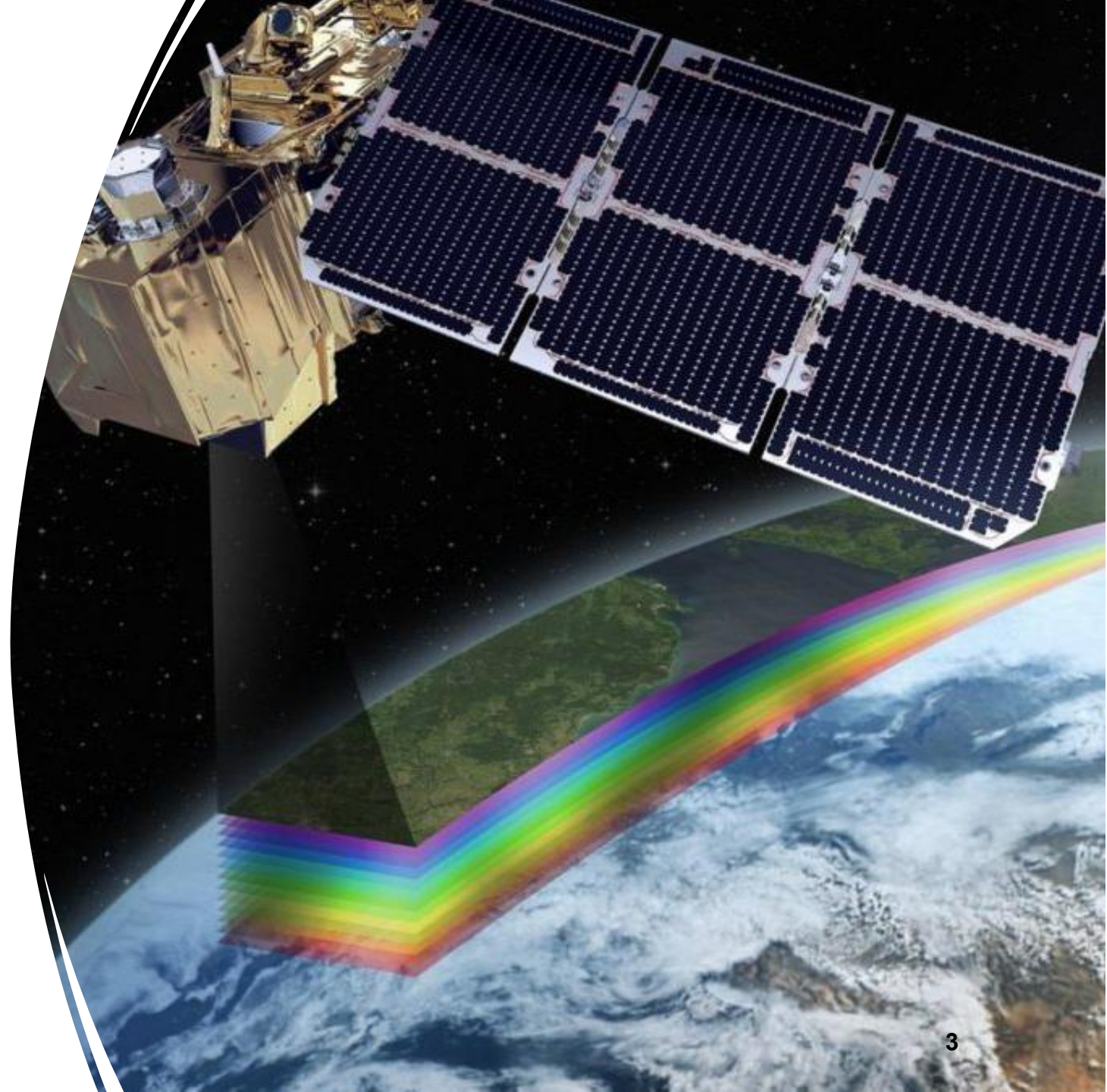
Motivation

- Wildfires are increasing due to climate change
- Wildfires emit pollutants that affect human health
- Understanding pollution episodes helps improve forecasting and decision making
- Wildfire emissions are becoming an increasing threat to air quality across Southern Europe



Objectives

- Investigate the relationship between wildfires and air pollution
- Evaluate the influence of meteorological conditions
- Detect pollution events using remote sensing
- Develop an LSTM model to predict pollutant concentrations



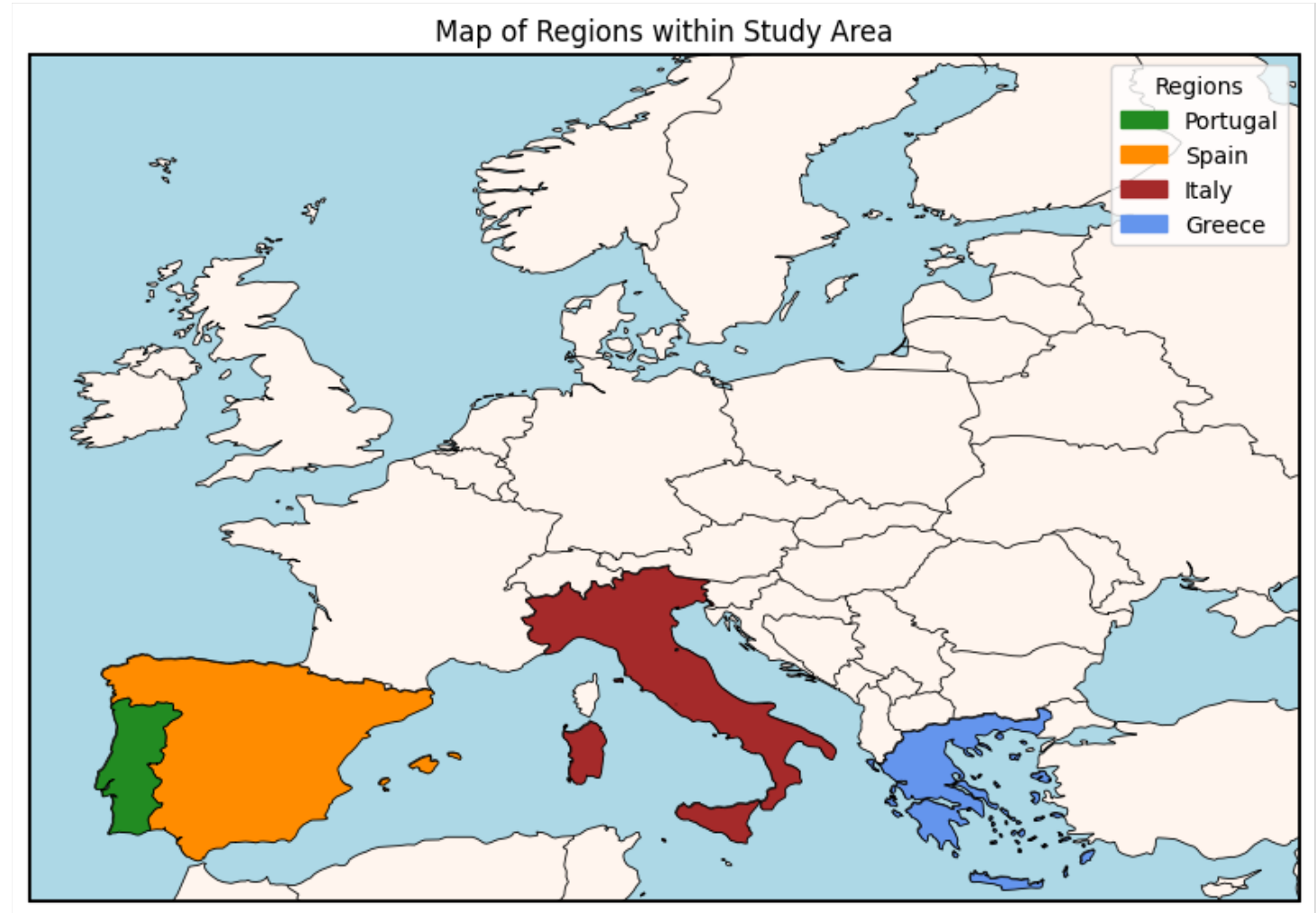
Study Area

Regions studied:

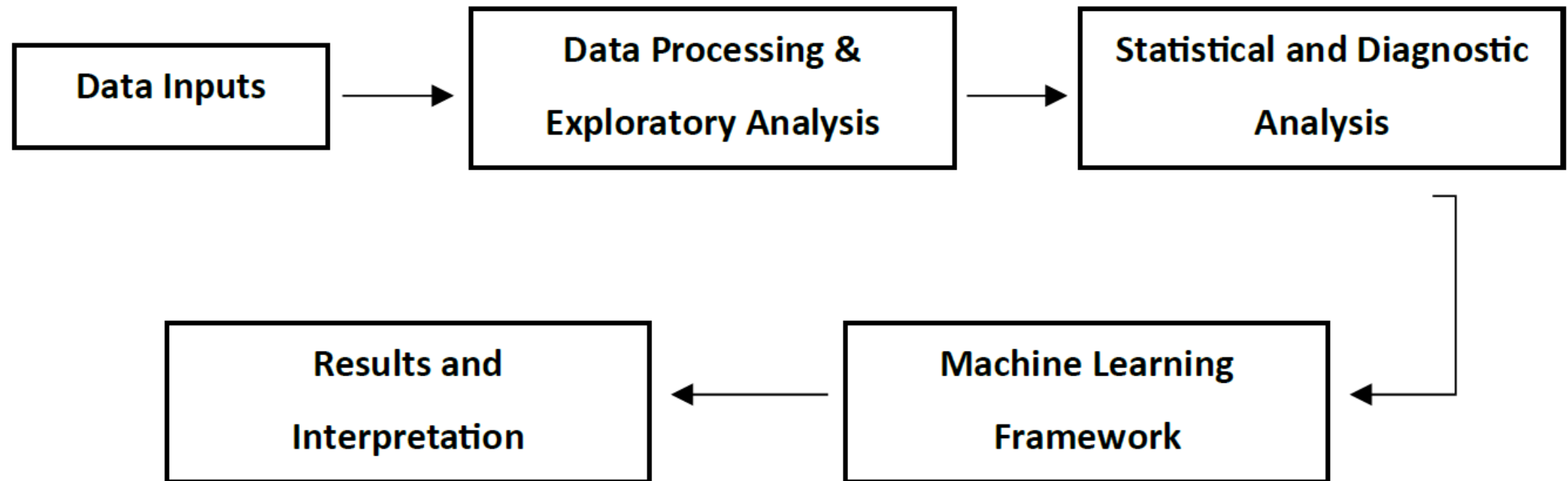
- Iberia – Portugal (2003, 2005, 2017, 2024) and Spain (2003, 2006, 2012, 2021, 2022)
- Italy (2007, 2012, 2017, 2021, 2023)
- Greece (2007, 2021, 2023)

Study period:

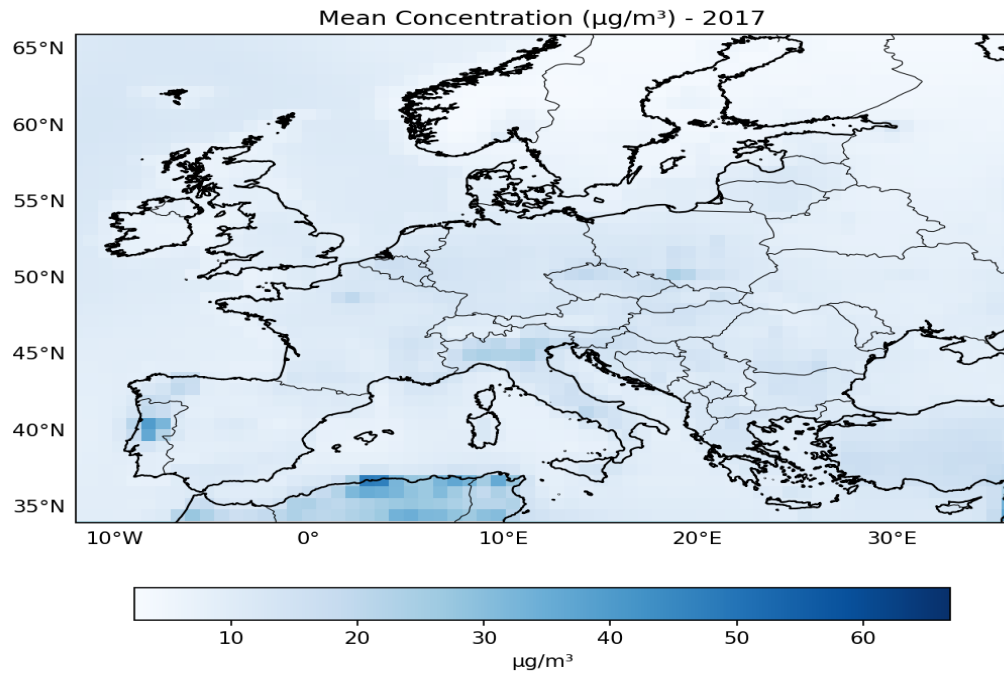
- 2003-2024



Methodology

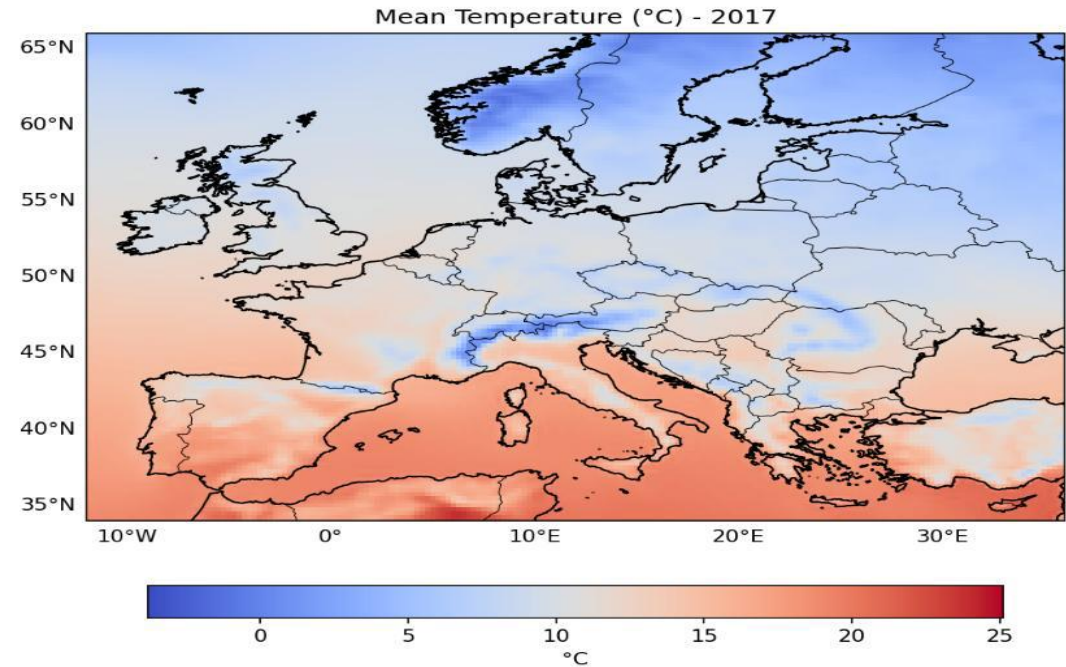


Data



**CAMS-EAC4 – CO, NO, NO₂,
PM_{2.5} and PM₁₀**

Atmosphere Data Store



**ERA5 – Temperature, Wind
and Precipitation**

Climate Data Store

Data

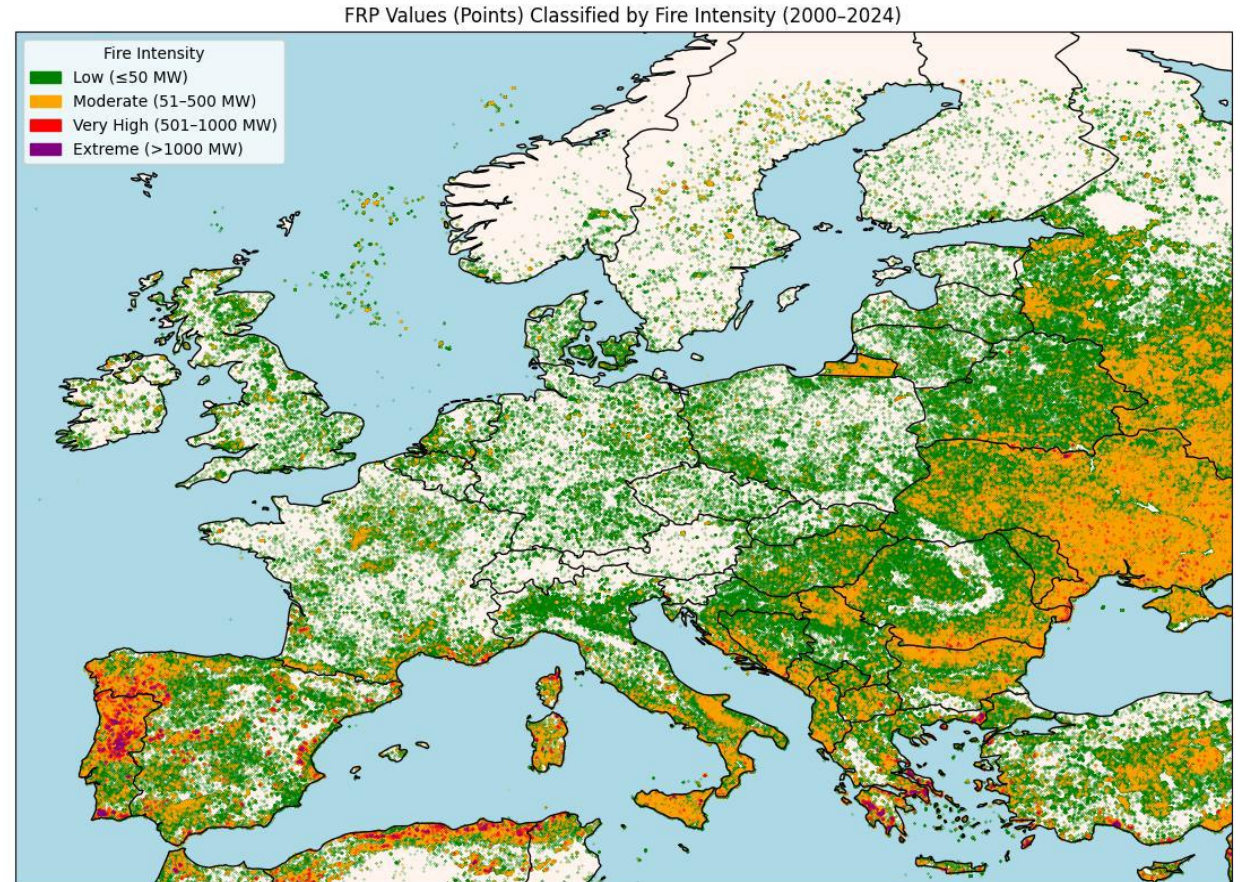
Fire Radiative Power (FRP)

NASA (Aqua and Terra satellites)

First Objectives:

Common grid: $0.75^\circ \times 0.75^\circ$

Daily Statistics



Data Manipulation

Masking of all regions in WGS84, spatial resolution $0.75^{\circ} \times 0.75^{\circ}$

Unit Conversion for Meteorological and Pollutant Data | Wind speed and direction calculation

Transform Point Data into Grid Data (FRP)

Regrid of Meteorological Data from 0.25° to 0.75°

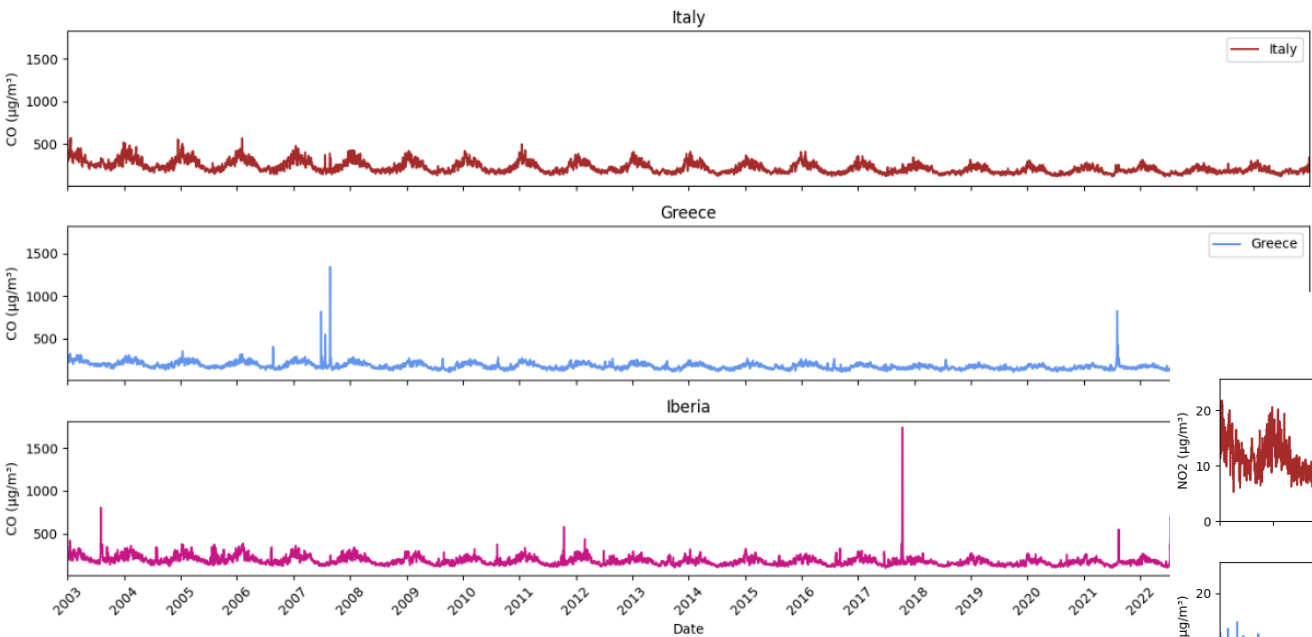
Model Development

Test year 2017 for Iberia

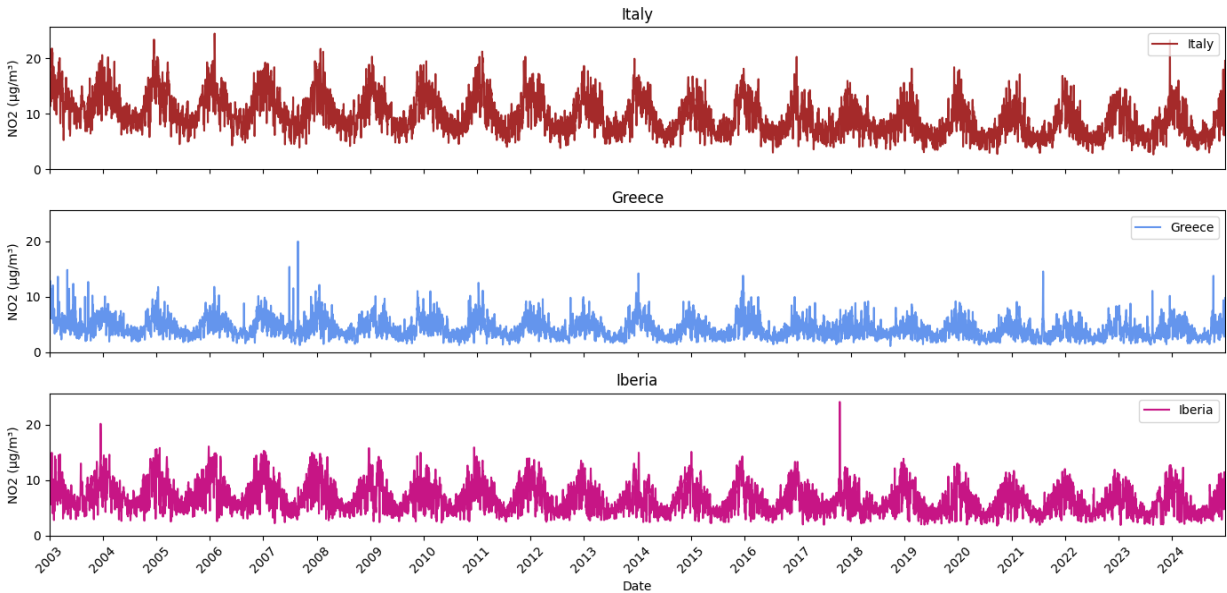
Predictor Group	Variables	Single-pollutant model	Multi-pollutant model	Persistence model
Target pollutant field	Mean concentration of the target pollutant	Yes	Yes	Yes (previous-day value only)
Other pollutant field	CO, NO, NO ₂ , PM _{2.5} , PM ₁₀ daily mean concentrations	No	Yes	No
Meteorology	Maximum temperature, maximum wind speed, total precipitation	Yes	Yes	No
Wildfire activity	Daily sum of FRP over Iberia	Yes	Yes	No
CAMS pollutant variables	Daily 00h value and previous-day maximum for each pollutant	Yes (pollutant-specific)	Yes (all pollutants)	No

Exploratory Analysis

Daily Mean CO Concentration ($\mu\text{g}/\text{m}^3$) by Region

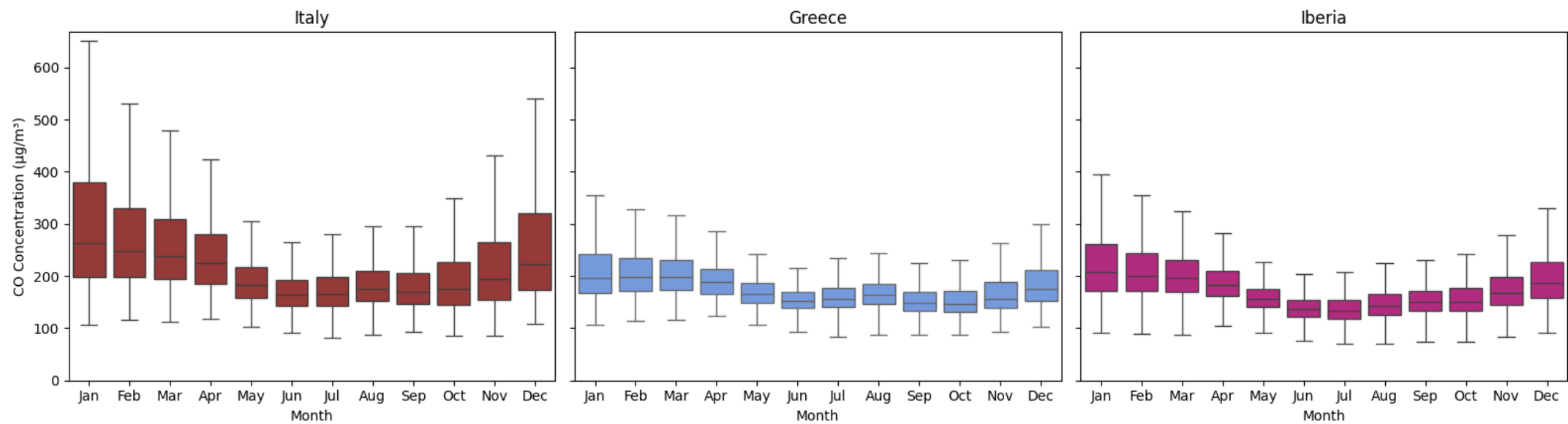


Daily Mean NO2 Concentration ($\mu\text{g}/\text{m}^3$) by Region



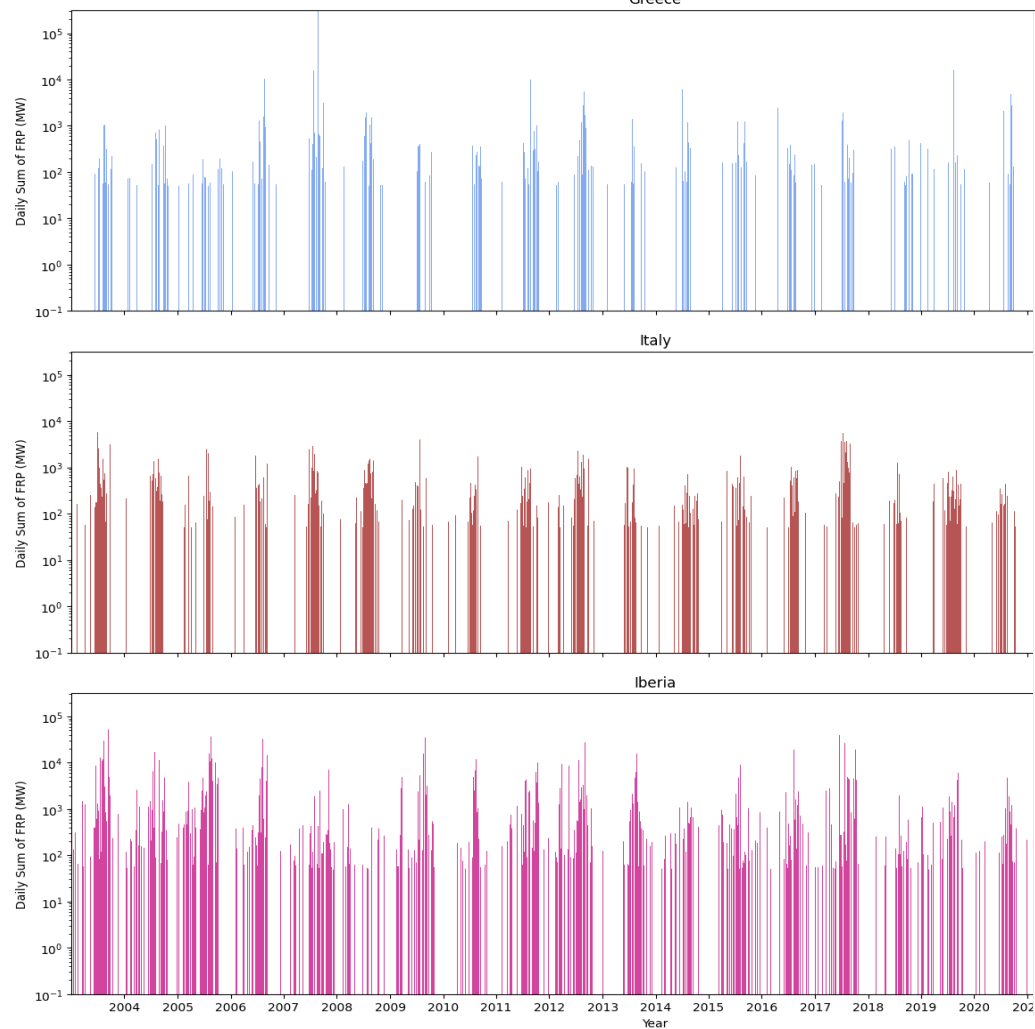
Exploratory Analysis

Monthly Variability of Daily Mean CO Concentration by Region

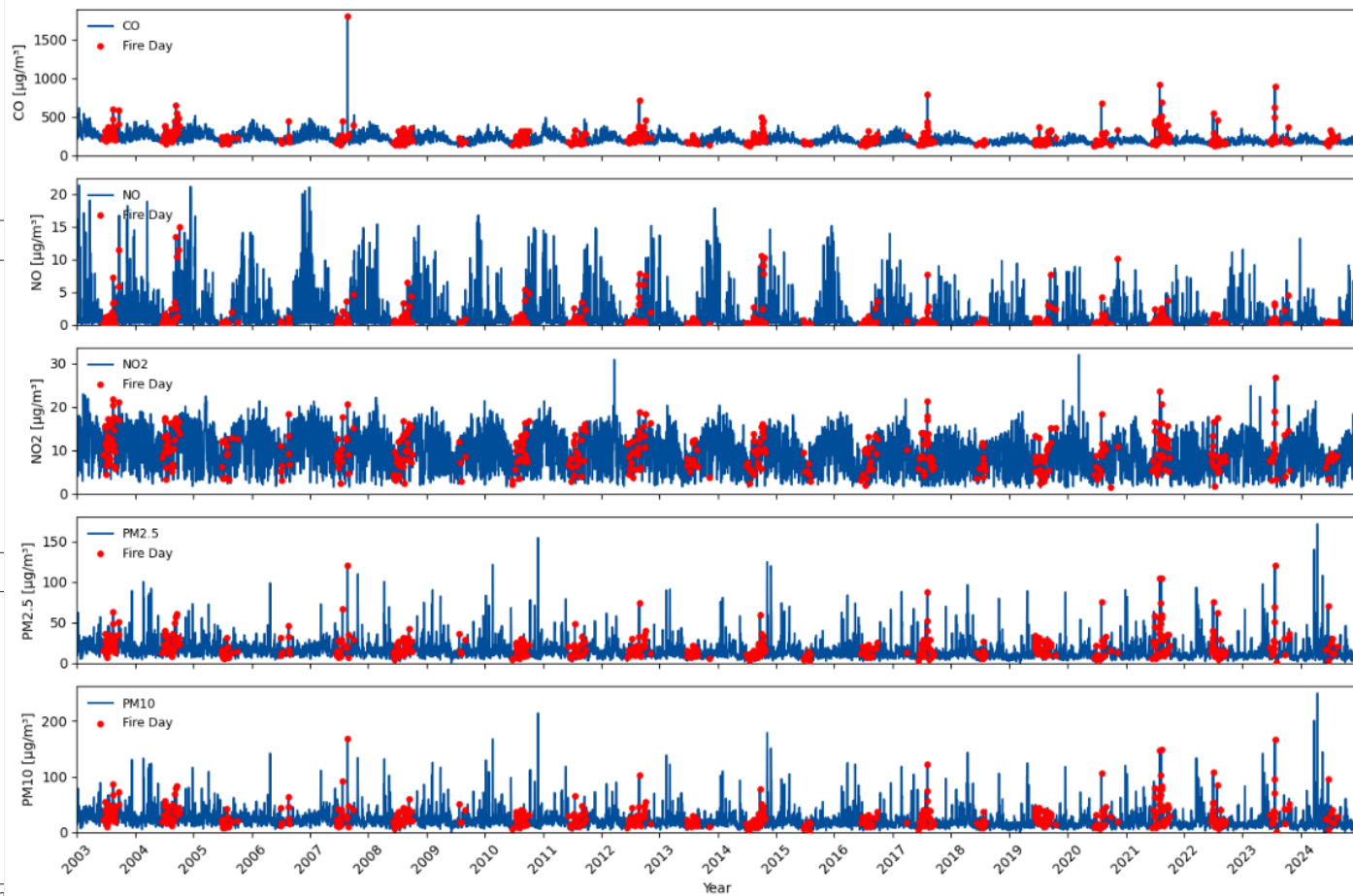


Exploratory Analysis

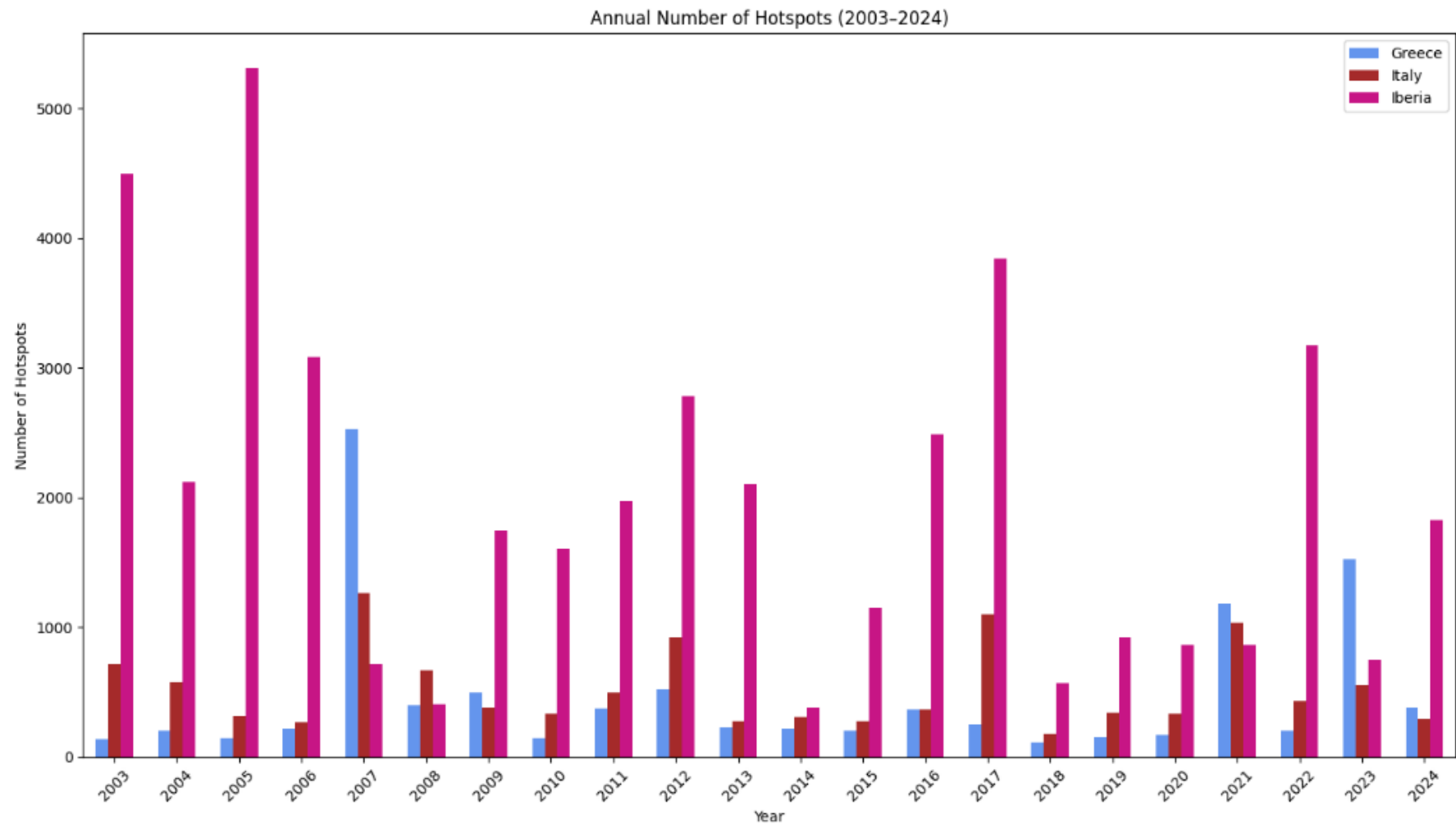
Daily Sum of FRP by Region from 2003 to 2024 (Log Scale)



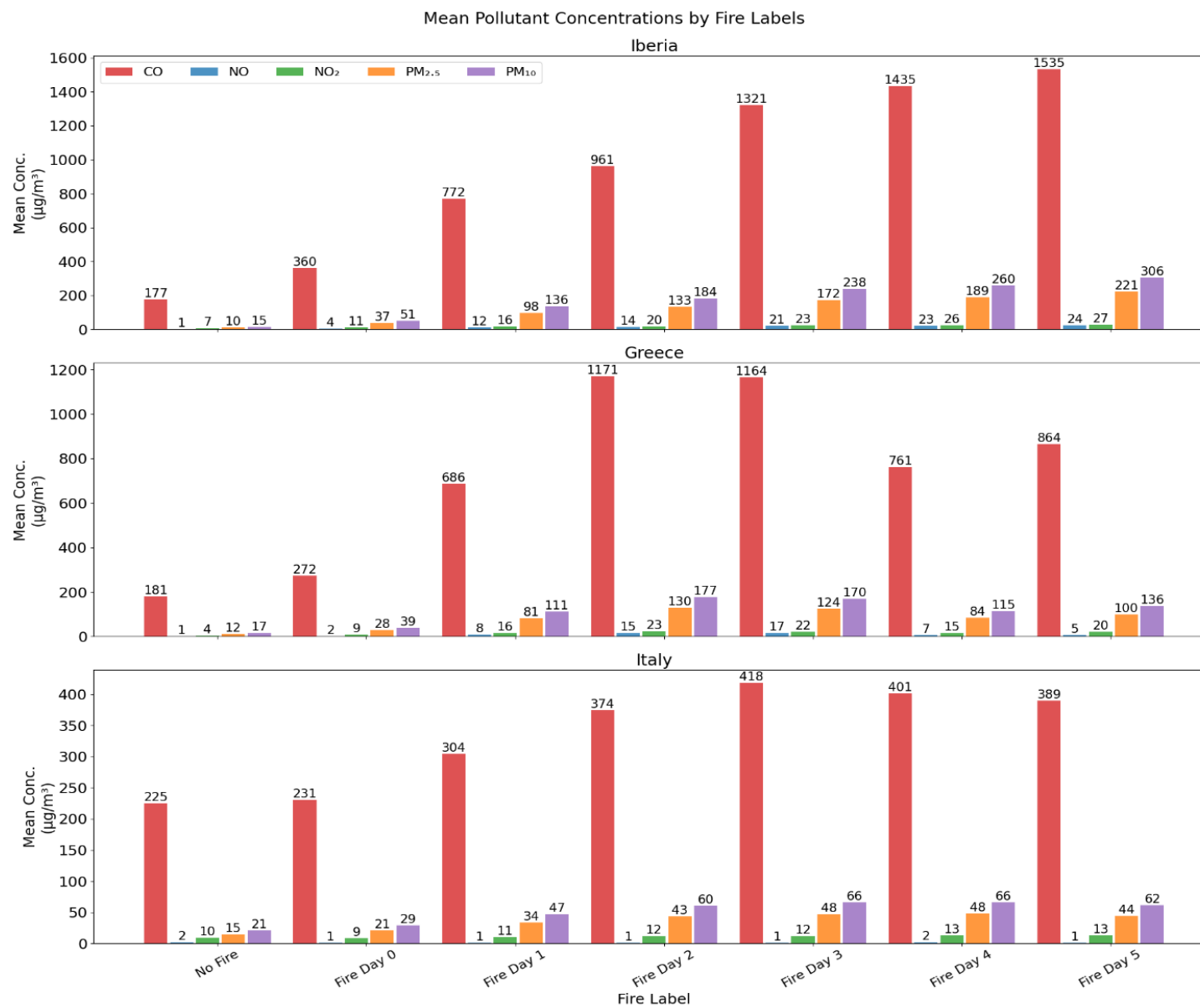
Pollutant Concentrations at Pixel with Most Fire Days (37.500, 14.250) — Italy



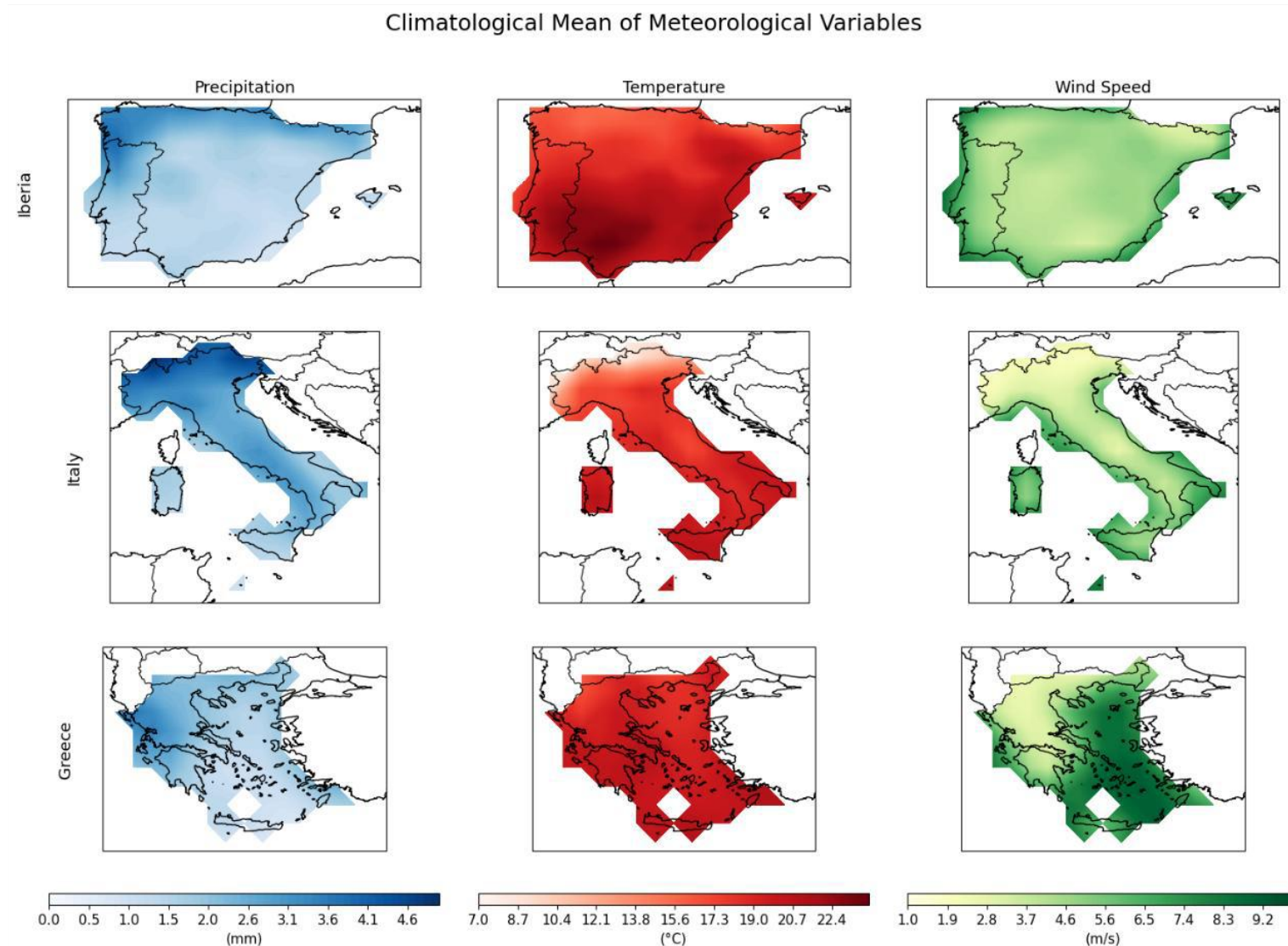
Exploratory Analysis



Fire Labelling

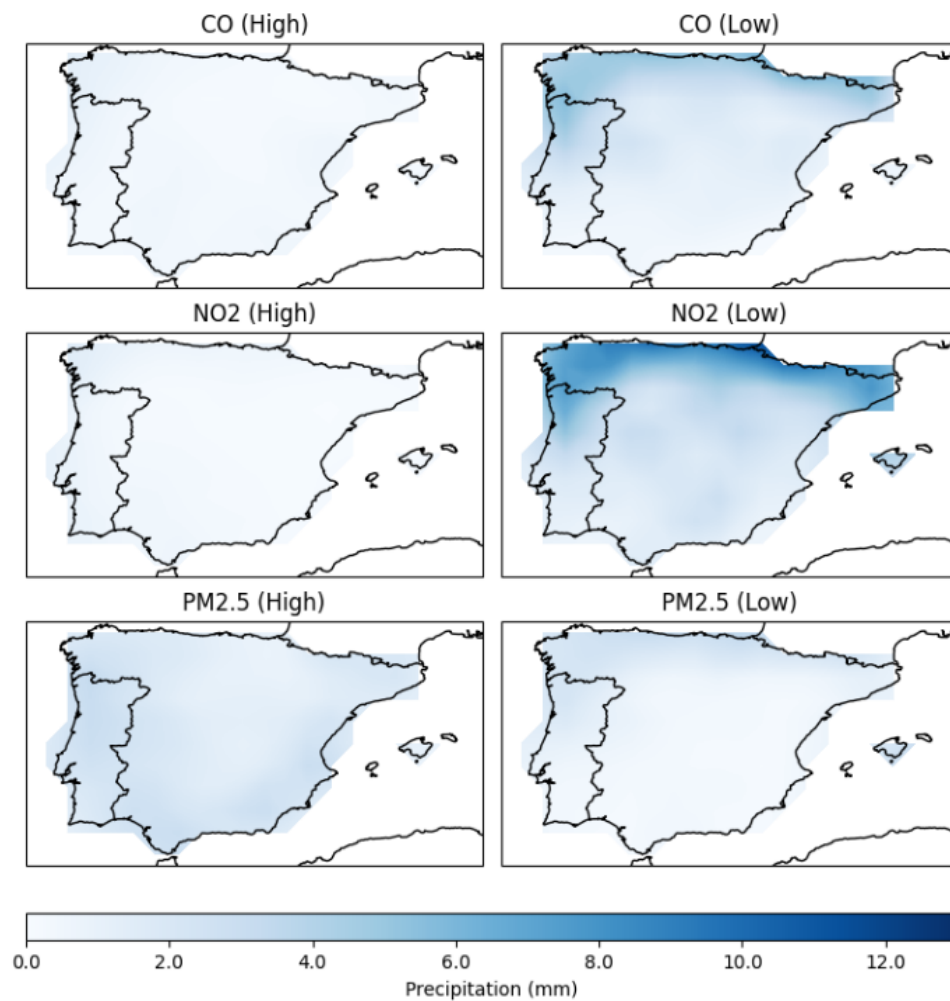


Annual Climatological Spatial Patterns over 22 years

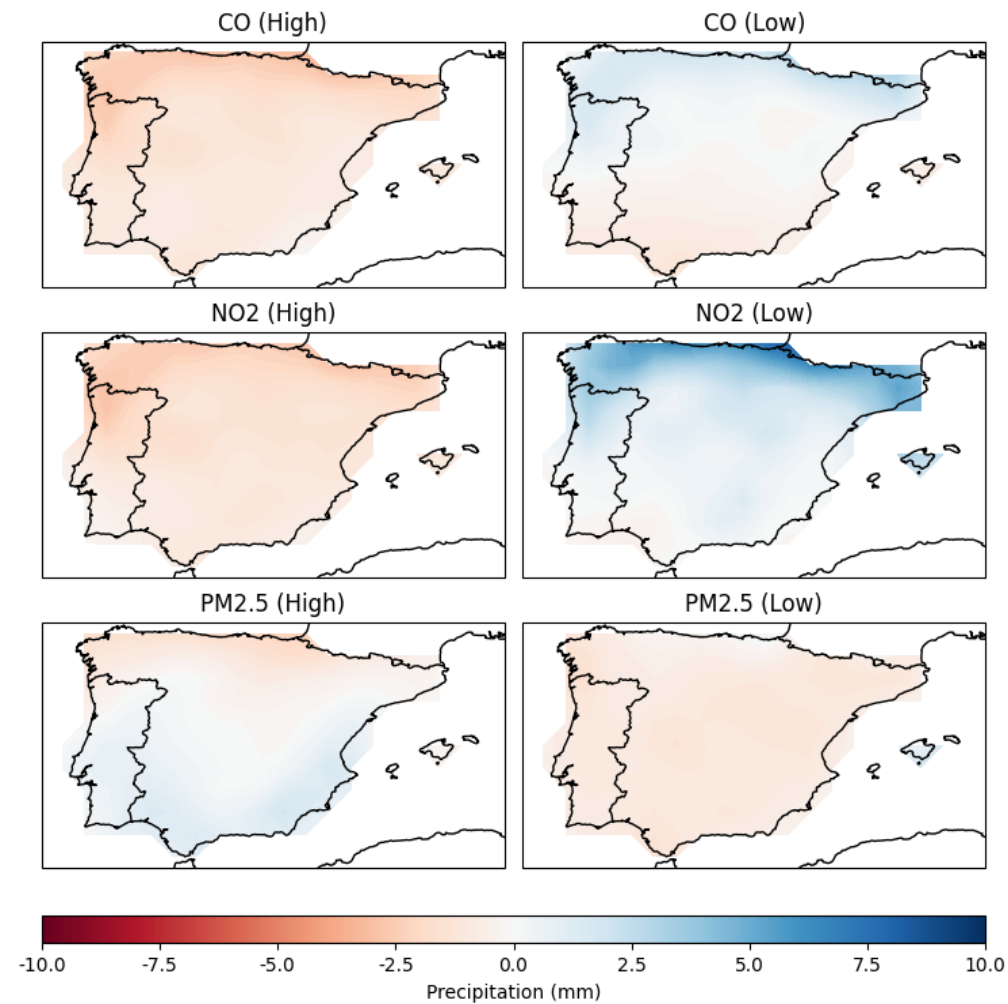


Composites and Anomalies

Precipitation — High and Low Pollutant Composites



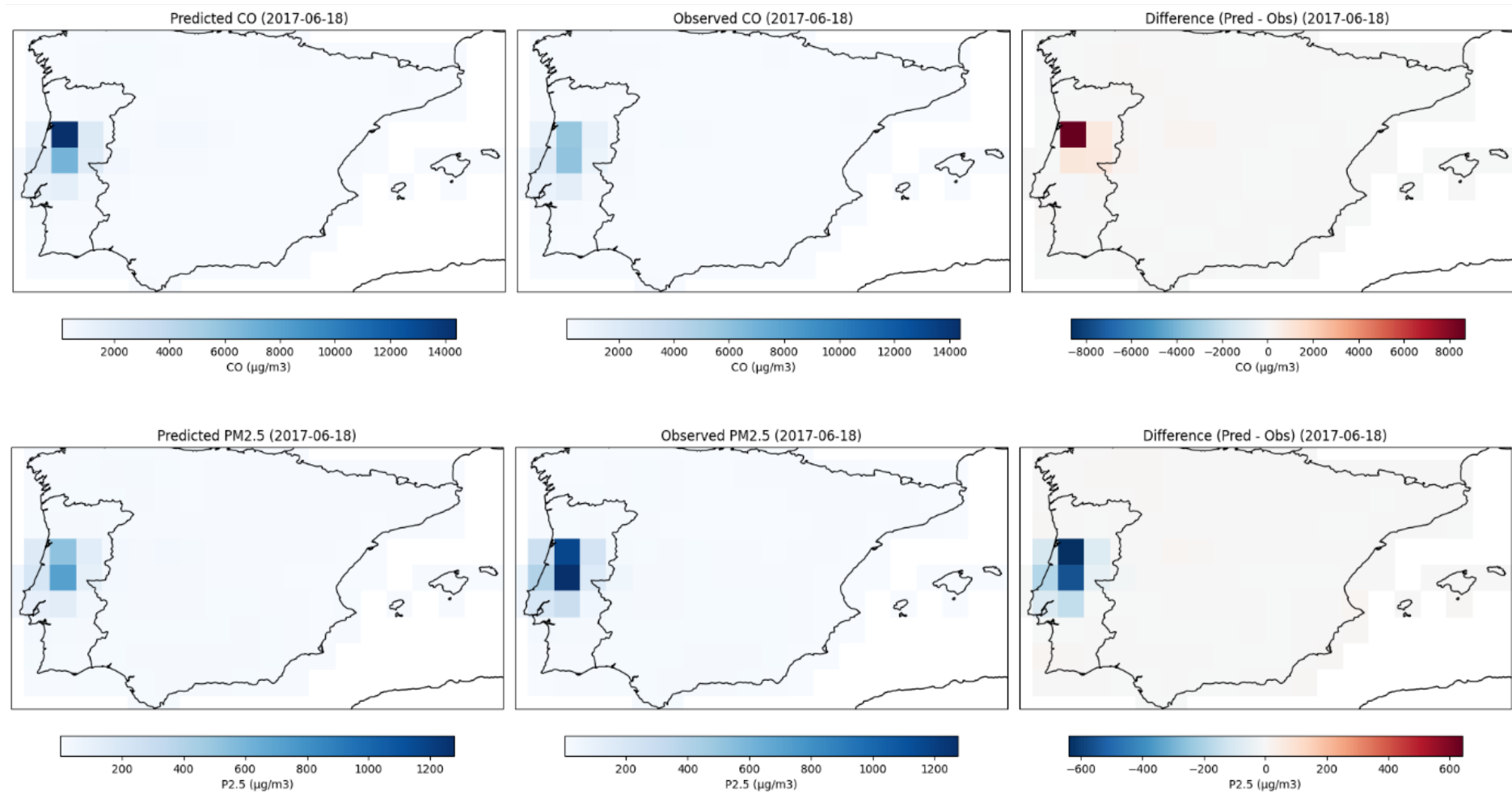
Precipitation — High and Low Pollutant Anomalies



Model Predictions

Pollutants			CO ($\mu\text{g}/\text{m}^3$)	NO ($\mu\text{g}/\text{m}^3$)	NO ₂ ($\mu\text{g}/\text{m}^3$)	PM _{2.5} ($\mu\text{g}/\text{m}^3$)	PM ₁₀ ($\mu\text{g}/\text{m}^3$)
MAE	Single-pollutant model	with CAMS	0.0249	0.7288	1.6153	2.9828	4.1446
		without CAMS	0.0361	1.1254	1.7274	4.9395	6.9589
	Multi-pollutant model	with CAMS	0.0256	0.6848	1.5920	3.3184	4.7190
		without CAMS	0.0346	0.8647	1.7301	4.7432	7.0055
	Persistence model	with CAMS	0.0357	0.9522	2.1263	5.1718	7.2995
RMSE	Single-pollutant model	with CAMS	0.3255	8.2961	3.5757	15.6437	19.6969
		without CAMS	0.4160	9.7025	4.5568	24.5928	32.3548
	Multi-pollutant model	with CAMS	0.2752	8.1153	3.4154	15.1392	24.1027
		without CAMS	0.4444	9.9676	4.8081	23.9650	33.2617
	Persistence model	with CAMS	0.5083	12.2472	5.4980	27.3695	36.6814
Bias	Single-pollutant model	with CAMS	-0.0051	-0.2245	0.2683	-1.2279	-0.1921
		without CAMS	0.0038	0.3871	-0.0086	-1.6938	-0.0255
	Multi-pollutant model	with CAMS	0.0005	-0.1741	0.0478	-0.4957	-1.1537
		without CAMS	0.0037	-0.2093	0.1014	-0.3349	0.0223
	Persistence model	with CAMS	0.0003	0.0102	0.0106	0.0118	0.0141
Correlation	Single-pollutant model	with CAMS	0.8956	0.6919	0.7996	0.8824	0.8870
		without CAMS	0.7106	0.5132	0.6964	0.5975	0.5841
	Multi-pollutant model	with CAMS	0.8639	0.6938	0.7641	0.8311	0.8181
		without CAMS	0.7298	0.5321	0.6914	0.6157	0.5860
	Persistence model	with CAMS	0.3383	0.2681	0.5866	0.5633	0.5724

Model Predictions - Visualization



Discussion

How do FRP and meteorological variables correlate with measured air pollutant concentrations?

Can machine learning models accurately predict wildfire-induced pollution events using remote sensing and meteorological data?

What spatial or temporal patterns emerge from wildfire emissions in different regions?

Conclusion

Wildfires increase atmospheric pollutant concentrations

Temperature, precipitation and wind, strongly influence pollution episodes

LSTM performance allows to successfully predict daily pollutant concentrations

Integrating remote sensing and meteorological data improves wildfire pollution detection

Improvements | Future works

Additional meteorological variables and other pollutants (e.g. Boundary Layer Height, Wind Direction, Ozone)

Compare LSTM model performance with others (e.g. Random Forest, XGBoost) and use SHAP to get insights on variables contribution

Improve uniformization of Spatial Data

The background of the slide is a photograph of a dense forest of evergreen trees, likely spruce or fir, with a strong blue color cast. The trees are layered, creating a sense of depth. The text is centered over the image.

Thank you for your attention!

Any Questions?